

HUNDRED DAYS OF CODE : DAY - 6

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I. Problem Statement - I

Write a C program to find the area using Heron's formula, semi-perimeter, and perimeter of a triangle when its three sides are given.

I.1 Inputs, Outputs & Formula Used

The Formula used for this problem is : $A = \sqrt{s(s-a)(s-b)(s-c)}$.
A = area, $s = \frac{1}{2}(a+b+c)$,
a,b,c = Sides of the triangle

The overview of the program is given as :

Test Case	Input	Expected Output	Actual Output	Result
1.	5,5,5	15 7.50 10.83	15 7.50 10.83	PASS
2.	4,4,4	14 7.0 7.94	14 7.0 7.94	PASS
3.	10,20,5	35 17.50 no area	35 17.50 no area	PASS

I.2 The algorithm

- I. Start
2. Declare variables side1, side2, side3, perimeter, semi_perimeter, area
3. Display "Enter the three sides of the triangle"
4. Read side1, side2, side3 from user
5. Check if (side1 <= 0 OR side2 <= 0 OR side3 <= 0)
 If YES → Display "Error: All sides must be positive" Go to Step II
6. Check if (side1 + side2 <= side3 OR side1 + side3 <= side2 OR side2 + side3 <= side1)
 If YES → Display "Error: Invalid triangle"
 Go to Step II
7. Calculate perimeter = side1 + side2 + side3
8. Calculate semi_perimeter = perimeter / 2
9. Calculate area = $\sqrt{\text{semi_perimeter} * (\text{semi_perimeter} - \text{side1}) * (\text{semi_perimeter} - \text{side2}) * (\text{semi_perimeter} - \text{side3})}$
10. Display perimeter, semi_perimeter, area
- II. Stop

1.3 The C Program

```
#include <stdio.h>
#include <math.h>

int main()
{
    float a, b, c, perimeter, s, area;

    printf("Enter three sides of the triangle: ");
    scanf("%f %f %f", &a, &b, &c);

    perimeter = a + b + c;
    s = perimeter / 2.0;
    area = sqrt(s * (s - a) * (s - b) * (s - c));

    printf("Perimeter: %.2f\n", perimeter);
    printf("Semi-perimeter: %.2f\n", s);
    printf("Area: %.2f\n", area);

    return 0;
}
```

I.4 Output

```
(kali㉿kali)-[~/codes/day_6]
└─$ ls
que1.c

(kali㉿kali)-[~/codes/day_6]
└─$ gcc que1.c -o que1 -lm

(kali㉿kali)-[~/codes/day_6]
└─$ ./que1
Enter three sides of the triangle: 5 5 5
Perimeter: 15.00
Semi-perimeter: 7.50
Area: 10.83

(kali㉿kali)-[~/codes/day_6]
└─$ 4 5 6
4: command not found

(kali㉿kali)-[~/codes/day_6]
└─$ ./que1
Enter three sides of the triangle: 4 4 6
Perimeter: 14.00
Semi-perimeter: 7.00
Area: 7.94

(kali㉿kali)-[~/codes/day_6]
└─$ 10 20 5
10: command not found

(kali㉿kali)-[~/codes/day_6]
└─$ ./que1
Enter three sides of the triangle: 10 20 5
Perimeter: 35.00
Semi-perimeter: 17.50
Area: -nan
```

2.Problem Statement - II

Write a C program to find the side, perimeter, and area of a regular hexagon when its perimeter is given.

2.1 Inputs, Outputs & Formula Used

The Formulas used for this problem is : 1. $a = P/6.0$

$$2. A = 3 * \sqrt{3} / 2 * a * a$$

where P = perimeter of hexagon

a = side of hexagon

A = area of the hexagon

The overview of the program is given as :

Test Case	Input	Expected Output	Actual Output	Result
1.	24	a: 4.00, P: 24.00, A: 41.57	a: 4.00, P: 24.00, A: 41.57	PASS
2.	36	a: 6.00, P: 36.00, A: 93.53	a: 6.00, P: 36.00, A: 93.53	PASS
3.	12	a: 2.00, P: 12.00, A: 10.39	a: 2.00, P: 12.00, A: 10.39	PASS

2.2 The algorithm

1. Start the program.
2. Declare variables for perimeter, side, and area.
3. Read the perimeter from the user.
4. Calculate $a = P / 6.0$.
5. Calculate $A = (3.0 * \sqrt{3} / 2.0) * a * a$.
6. Print a, P, and A.
7. Stop.

2.3 The C Program

```
#include <stdio.h>
#include <math.h>

int main()
{
    float p, side, area;

    printf("Enter perimeter of the regular hexagon: ");
    scanf("%f", &p);

    side = p / 6.0;
    area = (3.0 * sqrt(3.0) / 2.0) * side * side;

    printf("Side: %.2f\n", side);
    printf("Perimeter: %.2f\n", p);
    printf("Area: %.2f\n", area);

    return 0;
}
```

2.4 Output

```
(kali@kali)-[~/codes/day_6]
$ gedit que2.c

** (gedit:18976): WARNING **: 13:20:41.578: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:18976): WARNING **: 13:20:41.578: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

(kali@kali)-[~/codes/day_6]
$ gcc que2.c -o que2 -lm

(kali@kali)-[~/codes/day_6]
$ ./que2
Enter perimeter of the regular hexagon: 24
Side: 4.00
Perimeter: 24.00
Area: 41.57

(kali@kali)-[~/codes/day_6]
$ ./que2
Enter perimeter of the regular hexagon: 36
Side: 6.00
Perimeter: 36.00
Area: 93.53

(kali@kali)-[~/codes/day_6]
$ ./que2
Enter perimeter of the regular hexagon: 12
Side: 2.00
Perimeter: 12.00
Area: 10.39
```
